Quantum Boltzmann Machines and Summary of Course

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Outline

From Discriminative to Generative Models

Discriminative QNN (what we had)

- Input x encoded via $U(x)$
- Output: class label or regression value
- Minimise: $\mathcal{L} = \sum_k (f(x_k) - y_k)^2$
- Goal: learn $p(y|x)$

Generative QML (our goal now)

- **No** input encoding needed
- Output: full probability distribution $p(x)$
- Minimise: $D_{\mathrm{KL}}(p_{\mathrm{data}} \,\|\, p_\theta)$
- Goal: learn to *sample* from p_{data}

Generative models in QML:

- Quantum Boltzmann Machines (QBM) — *this talk*
- Quantum Circuit Born Machines (QCBM)
- Quantum GANs
- Quantum Variational Autoencoders

Key difference:
QBMs model distributions via a *thermal state* (density matrix) rather than a pure variational state.

Classical Restricted Boltzmann Machine

An RBM has **visible** units $\mathbf{v} \in \{0,1\}^n$ and **hidden** units $\mathbf{h} \in \{0,1\}^m$.

Energy function:

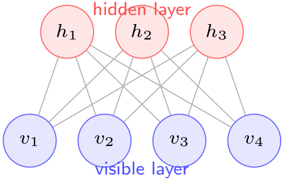
$$E(\mathbf{v},\mathbf{h}) = -\sum_i a_i v_i - \sum_j b_j h_j - \sum_{ij} v_i W_{ij} h_j$$

Joint distribution and partition function:

$$p(\mathbf{v},\mathbf{h}) = \frac{e^{-\beta E}}{Z}, \quad Z = \sum_{\mathbf{v},\mathbf{h}} e^{-\beta E}$$

Marginal over visible:

$$p(\mathbf{v}) = \frac{1}{Z} \sum_{\mathbf{h}} e^{-\beta E(\mathbf{v},\mathbf{h})}$$



No intra-layer connections $\Rightarrow$ tractable conditional independence:

$$p(h_j = 1|\mathbf{v}) = \sigma\big(b_j + \sum_i W_{ij} v_i\big)$$

RBM Training: Contrastive Divergence

Log-likelihood gradient:

$$\frac{\partial \ln p(\mathbf{v})}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}$$

- Data term: easy (one forward pass)
- Model term: hard (requires summing over all $\mathbf{v}, \mathbf{h}$)

Contrastive Divergence (CD-k):

1. Clamp $\mathbf{v}^{(0)} = \mathbf{v}_{\text{data}}$
2. Run k steps of Gibbs sampling $\mathbf{v}^{(0)} \to \mathbf{h}^{(0)} \to \mathbf{v}^{(1)} \to \cdots \to \mathbf{v}^{(k)}$
3. Approximate model term with $\mathbf{v}^{(k)}$

CD-k update rule

$$\Delta W_{ij} \propto \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_k$$

Similarly for biases a_i, b_j.

Key limitation

The partition function Z is intractable for large systems — gradient estimates are biased. Quantum hardware promises an exact, unbiased estimate of $\langle v_i h_j \rangle_{\text{model}}$ via *quantum sampling*.

From Classical to Quantum Gibbs States

Replace binary units with **qubits** and the classical energy function with a **quantum Hamiltonian**:

Classical: $p(\mathbf{v}) = e^{-\beta E(\mathbf{v})}/Z$

Quantum: The thermal (Gibbs) state

$$\boxed{\rho(\theta) = \frac{e^{-\beta H(\theta)}}{\mathrm{Tr}\left[e^{-\beta H(\theta)}\right]}}$$

where $H(\theta)$ is a parametrised quantum Hamiltonian.

The partition function is now a *quantum trace*:

$$Z(\theta) = \mathrm{Tr}\left[e^{-\beta H(\theta)}\right] = \sum_k e^{-\beta \epsilon_k(\theta)}$$

where $\{\epsilon_k\}$ are eigenvalues of $H(\theta)$.

Degrees of freedom

Classical RBM	QBM
Binary spins $\{0,1\}^n$	n qubits
Energy $E(\mathbf{v},\mathbf{h})$	Hamiltonian $H(\theta)$
Gibbs dist. $e^{-\beta E}/Z$	Gibbs state $e^{-\beta H}/Z$
Marginal $p(\mathbf{v})$	Reduced state ρ_v

Key gain

Quantum off-diagonal (transverse) terms in $H(\theta)$ allow the QBM to represent *quantum correlations* and *superpositions* — beyond any classical BM.

The QBM Hamiltonian

The most general parametrised n-qubit QBM Hamiltonian:

$$H(\theta) = -\sum_i \theta_i^x X_i - \sum_i \theta_i^z Z_i - \sum_{i<j} \theta_{ij}^{zz} Z_i Z_j - \cdots$$

- θ_i^z: **longitudinal fields** (diagonal) — classical bias terms, analogous to a_i
- θ_{ij}^{zz}: **ZZ couplings** (diagonal) — classical two-body interactions, analogous to W_{ij}
- θ_i^x: **transverse fields** (off-diagonal) — purely quantum term, no classical analogue

Thermal state in the eigenbasis

$$H(\theta)\,|\epsilon_k\rangle = \epsilon_k\,|\epsilon_k\rangle$$

$$\rho(\theta) = \sum_k \frac{e^{-\beta\epsilon_k}}{Z}\,|\epsilon_k\rangle\langle\epsilon_k|$$

Classical limit

Setting $\theta_i^x = 0$ for all i recovers a classical BM: $[H, Z_i] = 0\ \forall i$, so ρ is diagonal in the Z-basis and reduces to a classical Boltzmann distribution.

Visible, Hidden, and Auxiliary Qubits

Following the classical RBM structure, the n qubits are partitioned into:

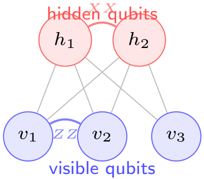
$$\mathcal{H} = \mathcal{H}_v \otimes \mathcal{H}_h, \quad n_v + n_h = n$$

- **Visible** qubits v: coupled to the data; we observe their reduced state $\rho_v = \mathrm{Tr}_h[\rho]$
- **Hidden** qubits h: latent variables that are traced out

The visible marginal is the model distribution:

$$p_\theta(\mathbf{v}) = \langle \mathbf{v} | \, \rho_v(\theta) \, | \mathbf{v} \rangle$$

where $\rho_v(\theta) = \mathrm{Tr}_h\left[\frac{e^{-\beta H(\theta)}}{Z}\right]$.



Quantum connections can include XX, YY, ZZ within visible, within hidden, and across layers.

The QBM Training Objective

Given empirical distribution $\pi(\mathbf{v})$, minimise the KL divergence $\mathcal{L}(\theta) = D_{\mathrm{KL}}(\pi \,\|\, p_\theta)$:

$$\mathcal{L}(\theta) = \sum_{\mathbf{v}} \pi(\mathbf{v}) \ln \frac{\pi(\mathbf{v})}{p_\theta(\mathbf{v})}$$

Expanding with $p_\theta(\mathbf{v}) = \langle \mathbf{v} | \rho_v | \mathbf{v} \rangle$ and $\sigma_{\mathrm{data}} = \sum_{\mathbf{v}} \pi(\mathbf{v}) | \mathbf{v} \rangle \langle \mathbf{v} |$:

$$\mathcal{L}(\theta) = -\operatorname{Tr}[\sigma_{\mathrm{data}} \ln \rho_v(\theta)] + \mathrm{const}$$

Quantum relative entropy

The objective equals the quantum relative entropy:

$$\mathcal{L} = S(\sigma_{\mathrm{data}} \,\|\, \rho_v(\theta))$$

$$= \operatorname{Tr}[\sigma_{\mathrm{data}}(\ln \sigma_{\mathrm{data}} - \ln \rho_v)]$$

Reduces to classical KL when both states are diagonal.

Full QBM objective

For a quantum target state, use the total quantum relative entropy: $\mathcal{L} = S(\rho_{\mathrm{target}} \,\|\, \rho(\theta))$.

Gradient of the Log-Likelihood

The gradient of the NLL w.r.t. θ_μ, where $H(\theta) = \sum_\mu \theta_\mu \Lambda_\mu$:

$$\frac{\partial \mathcal{L}}{\partial \theta_\mu} = \langle \Lambda_\mu \rangle_{\rho_{\text{clamp}}} - \langle \Lambda_\mu \rangle_{\rho(\theta)}$$

The two expectation values (Λ_μ = generalised force):

$$\begin{aligned} \langle \Lambda_\mu \rangle_{\text{clamp}} &= \mathrm{Tr}[\Lambda_\mu \, \rho_{\text{clamp}}] \text{ (data-dependent)} \\ \langle \Lambda_\mu \rangle_\theta &= \mathrm{Tr}[\Lambda_\mu \, \rho(\theta)] \text{ (model-dependent)} \end{aligned}$$

Clamped state

ρ_{clamp} = Gibbs state with visible qubits clamped to a data sample:

$$\rho_{\text{clamp}} = \frac{e^{-\beta H^{(v)}}}{Z^{(v)}}$$

$H^{(v)}$: $H(\theta)$ with visible operators replaced by classical values.

Classical analogy

Quantum analogue of:

$$\frac{\partial \ln p}{\partial W_{ij}} = \langle v_i h_j \rangle_{\text{data}} - \langle v_i h_j \rangle_{\text{model}}$$

The Quantum Log-Likelihood Gradient: Derivation

Key identity. For $H(\theta)$ depending smoothly on θ_μ:

$$\frac{\partial}{\partial \theta_\mu} \ln Z = -\beta \ \mathrm{Tr}[\Lambda_\mu \, \rho(\theta)] = -\beta \langle \Lambda_\mu \rangle_\theta$$

Using $p_\theta(\mathbf{v}) = \langle \mathbf{v} | \, \rho_v \, | \mathbf{v} \rangle$:

$$\frac{\partial}{\partial \theta_\mu} \sum_{\mathbf{v}} \pi(\mathbf{v}) \ln p_\theta(\mathbf{v}) = \sum_{\mathbf{v}} \pi(\mathbf{v}) \frac{\partial}{\partial \theta_\mu} \ln p_\theta(\mathbf{v})$$

After using the **Golden-Thompson inequality** and the **Duhamel formula** for matrix exponential derivatives:

$$\frac{\partial \mathcal{L}}{\partial \theta_\mu} = \beta \left[\langle \Lambda_\mu \rangle_{\mathrm{clamp}} - \langle \Lambda_\mu \rangle_\theta \right]$$

Physical interpretation

Each gradient step reduces the difference between the *clamped* (data-driven) and *free* (model) thermal averages of Λ_μ — the quantum generalisation of Contrastive Divergence.

Challenges: The Partition Function and Thermal Sampling

The intractability problem

Computing $\langle \Lambda_\mu \rangle_\theta = \mathrm{Tr}[\Lambda_\mu \rho(\theta)]$ requires $\rho(\theta) = e^{-\beta H(\theta)} / \mathrm{Tr}\left[e^{-\beta H(\theta)}\right]$:

- Diagonalising a $2^n \times 2^n$ matrix — exponential cost classically
- Computing the matrix exponential and its trace
- Repeated for every gradient step

Three main approaches:

1. Quantum hardware sampling
2. Variational approximation of $\rho(\theta)$
3. Semiclassical / mean-field approximations

Quantum hardware advantage

A quantum device *naturally* prepares thermal states via cooling / environment coupling. Measuring $\mathrm{Tr}[\Lambda_\mu \rho]$ requires only $O(\varepsilon^{-2})$ shots — no diagonalisation.

Variational approach

Approximate the Gibbs state:

$$\rho(\theta) \approx \rho_\phi = U(\phi)\rho_0 U^\dagger(\phi)$$

Jointly optimise (θ, ϕ) — the **Quantum-Classical Boltzmann Machine**.

Method 1: Quantum imaginary-time evolution (QITE)
Evolve with $e^{-\Delta\tau H}$ approximated by unitaries (TDVP):

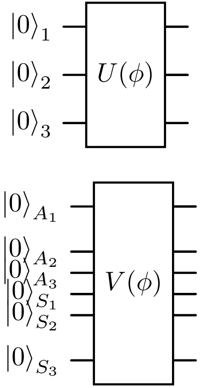
$$\rho(\tau) = \frac{e^{-\tau H}}{Z(\tau)}, \quad \tau = \beta/2$$

Each step: $e^{-\Delta\tau H}\left|\psi\right\rangle \approx U_{\mathrm{QITE}}\left|\psi\right\rangle$ where U_{QITE} solves $\sum_j F_{ij}\dot{\theta}_j = C_i$.

Method 2: Variational thermal ansatz

$$\rho_\phi = \mathrm{Tr}_A\left[U(\phi)\left|0\right\rangle_{SA}\left\langle 0\right| U^\dagger(\phi)\right]$$

Ancilla qubits A purify ρ via the thermofield double.



Top: direct variational state. Bottom: thermofield double (ancilla + system).

Purification and the Thermofield Double State

Any mixed state ρ_S can be **purified**: there exists $|\Psi\rangle_{SA}$ with $\rho_S = \mathrm{Tr}_A[|\Psi\rangle\langle\Psi|]$.
For the Gibbs state the canonical purification is the **thermofield double** (TFD):

$$|\mathrm{TFD}(\beta)\rangle = \frac{1}{\sqrt{Z(\beta)}} \sum_k e^{-\beta\epsilon_k/2} |\epsilon_k\rangle_S |\epsilon_k\rangle_A$$

Tracing out A recovers $\mathrm{Tr}_A[|\mathrm{TFD}\rangle\langle\mathrm{TFD}|] = \rho(\beta)$.

TFD as a QNN circuit

Prepare with a $2n$-qubit variational circuit:

$$|\mathrm{TFD}(\beta)\rangle \approx V(\phi)\,|0\rangle^{\otimes 2n}$$

$V(\phi)$ trained to minimise the free energy:

$$\mathcal{F}(\phi) = \langle H\rangle_\phi - \beta^{-1} S_{\mathrm{vN}}(\rho_\phi)$$

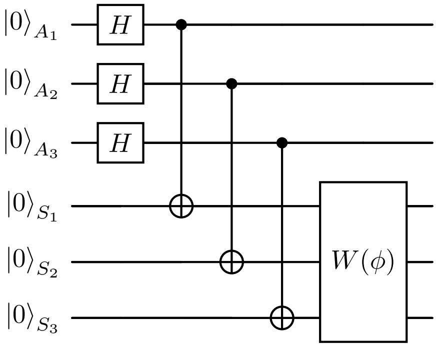
where $S_{\mathrm{vN}}(\rho) = -\mathrm{Tr}[\rho \ln \rho]$.

Key connection

Preparing the TFD $\equiv$ training a variational QNN to minimise the free energy of $H(\theta)$.

TFD Circuit: Quantikz Diagram

The thermofield double on $n = 3$ system $+ 3$ ancilla qubits:



- **Step 1:** $H^{\otimes n}$ on ancillas $\rightarrow$ uniform superposition
- **Step 2:** CNOT from each ancilla to its system partner $\rightarrow$ maximally entangled Bell pairs
- **Step 3:** $W(\phi)$ on system qubits tilts toward the correct thermal distribution
- Final: $\rho_S = \mathrm{Tr}_A[\cdot] \approx \rho(\beta)$

Variational QBM: Full Algorithm

The **Variational QBM** jointly optimises:

1. **Model params** θ: couplings in $H(\theta)$
2. **Circuit params** ϕ: prepares $\rho_\phi \approx \rho(\beta, \theta)$

Algorithm (one step):

1. Sample $\mathbf{v}$ from dataset
2. Run TFD circuit $V(\phi)$; get ρ_ϕ
3. Clamp visible qubits; get ρ_{clamp}
4. Estimate $\langle \Lambda_\mu \rangle_\phi$ and $\langle \Lambda_\mu \rangle_{\mathrm{clamp}}$
5. Update θ by gradient descent
6. Update ϕ by imaginary-time TDVP

Two nested optimisations

$$\min_\theta \ \mathcal{L}(\theta) = D_{\mathrm{KL}}(\pi \,\|\, p_\theta)$$

$$\min_\phi \ \mathcal{F}(\phi, \theta) = \langle H \rangle_\phi - \beta^{-1} S_{\mathrm{vN}}$$

Inner: ϕ tracks thermal state.
Outer: θ reduces KL.

Free-energy minimisation

$\mathcal{F}$ minimised when $\rho_\phi = \rho(\beta, \theta)$ — quantum analogue of the EM algorithm.

Free Energy and the QBM Objective

The **quantum free energy** at inverse temperature β:

$$\mathcal{F}(\rho, H) = \mathrm{Tr}[H\rho] - \beta^{-1} S_{\mathrm{vN}}(\rho)$$

$$S_{\mathrm{vN}}(\rho) = -\,\mathrm{Tr}[\rho \ln \rho]$$

The Gibbs state $\rho(\beta) = e^{-\beta H}/Z$ is the *unique* minimiser of $\mathcal{F}$:

$$\rho(\beta) = \arg\min_{\rho} \mathcal{F}(\rho, H)$$

Proof. Using the quantum relative entropy:

$$\mathcal{F}(\rho, H) = \mathcal{F}(\rho(\beta), H) + \beta^{-1} S(\rho \,\|\, \rho(\beta)) \geq \mathcal{F}(\rho(\beta), H)$$

since $S(\rho\|\sigma) \geq 0$ (Klein's inequality).

Variational free energy gradient

For a parametrised state ρ_ϕ:

$$\frac{\partial \mathcal{F}}{\partial \phi_k} = \mathrm{Tr}\left[H \frac{\partial \rho_\phi}{\partial \phi_k}\right] + \beta^{-1}\,\mathrm{Tr}\left[\ln \rho_\phi \frac{\partial \rho_\phi}{\partial \phi_k}\right]$$

The second term involves $\ln \rho_\phi$, which is expensive classically but accessible via Hadamard-test circuits on quantum hardware.

TDVP for imaginary time

The optimal update of ϕ at each step:

$$\sum_j F_{ij}^{(\phi)} \dot{\phi}_j = -\frac{\partial \mathcal{F}}{\partial \phi_i}$$

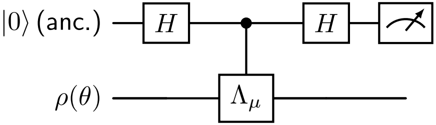
Gradient Computation: Quantum Circuit Approach

Evaluating $\langle \Lambda_\mu \rangle_\theta = \mathrm{Tr}[\Lambda_\mu \, \rho(\theta)]$:

1. Direct measurement (hardware):

1. Prepare $\rho(\theta)$ on the device
2. Measure the Pauli observable Λ_μ
3. Repeat N_{shots} times; estimate error $\sim 1/\sqrt{N_{\text{shots}}}$

2. Hadamard test (for off-diagonal Λ_μ):



$$\langle \Lambda_\mu \rangle_\theta = 2P(\text{ancilla} = 0) - 1$$

3. Parameter-shift for thermal states

For $\Lambda_\mu = P_\mu$ a Pauli operator:

$$\frac{\partial \langle O \rangle_\theta}{\partial \theta_\mu} = \frac{\beta}{2} \left[\langle O \rangle_{\theta_\mu + \delta} - \langle O \rangle_{\theta_\mu - \delta} \right]$$

with $\delta = \pi/(4r)$ where r is the eigenvalue of P_μ: for Paulis $r = 1/2$, so $\delta = \pi/2$.

Important caveat

This thermal parameter-shift rule requires that $\rho(\theta)$ can be re-prepared efficiently for shifted θ — true if the Gibbs state is prepared by a short circuit.

QBM as a Special QNN: Unified View

Both QNNs and QBMs manipulate parameterised quantum states, but the *state type* differs:

Feature	QNN	QBM
State	Pure $\vert\psi(\theta)\rangle$	Mixed $\rho(\theta)$
Objective	Supervised loss	KL divergence
Gradient	Param-shift	Thermal averages
Key tool	QFI / TDVP	Free energy / TDVP
Sampling	Projective	Thermal / Gibbs
Circuits	Unitary	Unitary + trace

Unifying framework

Both are instances of:

$$\min_{\theta} D(\sigma_{\text{target}} \,\|\, \mathcal{E}_\theta(\rho_0))$$

where $\mathcal{E}_\theta$ is a quantum channel (unitary for QNNs, Gibbs map for QBMs) and D is a divergence.

Advantage of QBMs

Mixed-state model $\Rightarrow$ can represent *classically correlated* distributions efficiently, without needing exponentially deep circuits to replicate thermal noise.

Connection to Many-Body Physics

The QBM Hamiltonian lives in the same world as condensed-matter and nuclear physics:

$$H(\theta) = -\sum_{i<j} \theta_{ij}^{zz} Z_i Z_j - \sum_i \theta_i^x X_i - \sum_i \theta_i^z Z_i$$

This is the transverse-field Ising model!

- θ_{ij}^{zz}: spin–spin coupling (Ising)
- θ_i^x: transverse field (quantum fluctuations)
- θ_i^z: longitudinal field (external bias)

Thermal state $\rho(\beta) = e^{-\beta H}/Z$ is the *exact* quantum statistical mechanics object studied in condensed matter — QBM training is parameter estimation for a quantum spin model.

Correspondence table

QBM	Many-body physics
Visible qubits	Measured spins
Hidden qubits	Auxiliary fields
θ_i^x	Transverse field
θ_{ij}^{zz}	Ising coupling
$\rho(\beta)$	Thermal density matrix
Training	Inverse problem

Inverse problem

Given samples from $\rho(\beta, \theta^\star)$, recover $\theta^\star$ — equivalent to learning the couplings of a quantum spin Hamiltonian from measurement data.

QBM and the Variational Principle

The QBM free-energy objective is an **upper bound on the free energy** of $H(\theta)$:

$$\mathcal{F}(\rho_\phi, H(\theta)) = \mathrm{Tr}[H(\theta)\rho_\phi] - \beta^{-1} S_{\mathrm{vN}}(\rho_\phi) \geq \mathcal{F}_{\mathrm{exact}}$$

Connection to VQE: VQE ($\beta \to \infty$) minimises $\langle H \rangle_\psi$ over pure states; QBM minimises $\mathcal{F}(\rho, H)$ over mixed states (finite T). Schematically:

$$\min_\psi \langle \psi | H | \psi \rangle \ \subset \ \min_\rho \mathcal{F}(\rho, H)$$

TDVP connection

The thermal gradient for ϕ:

$$\sum_j F^{(\phi)}_{ij} \dot{\phi}_j = -\partial_{\phi_i} \mathcal{F}$$

is the *imaginary-time TDVP* used in Stochastic Reconfiguration (SR).

QBM training $\equiv$ SR at finite temperature.

$\beta \to \infty$ limit

$\rho(\beta) \to |E_0\rangle \langle E_0|$, so QBM $\to$ VQE: the objective reduces to pure-state variational minimisation.

Quantum Natural Gradient for QBMs

Optimal update: $\Delta\phi = -\eta F^{-1}(\phi)\nabla_\phi \mathcal{F}$, where the **mixed-state QFI** (Bures metric) is:

$$F_{ij}^{(\phi)} = \mathrm{Tr}[\rho_\phi \mathcal{L}_i \mathcal{L}_j]$$

with symmetric logarithmic derivative $\mathcal{L}_i$ defined by:

$$\frac{\partial \rho_\phi}{\partial \phi_i} = \tfrac{1}{2}(\rho_\phi \mathcal{L}_i + \mathcal{L}_i \rho_\phi)$$

This generalises the pure-state QFI (week 15).

Bures vs. Fubini-Study metric

State	Metric
Pure $\lvert\psi\rangle$	Fubini-Study $F_{ij}^{\mathrm{FS}} = 4\,\mathrm{Re}(\dots)$
Mixed ρ	Bures $F_{ij}^{\mathrm{B}} = \mathrm{Tr}[\rho \mathcal{L}_i \mathcal{L}_j]$

Bures $\rightarrow$ Fubini-Study when $\rho = \lvert\psi\rangle\langle\psi\rvert$.

Practical computation

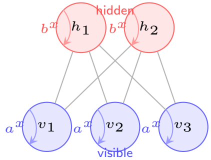
F_{ij}^{B} requires the SLD $\mathcal{L}_i$: expensive classically, but estimable via Hadamard-test circuits.

Restricted QBM: Architecture and Hamiltonian

The **Restricted QBM (RQBM)** is the direct quantum analogue of the classical RBM, with n_v visible and n_h hidden qubits:

$$H_{\mathrm{RQBM}} = -\sum_i a_i^z Z_i^v - \sum_j b_j^z Z_j^h - \sum_{ij} W_{ij} Z_i^v Z_j^h - \sum_i a_i^x X_i^v - \sum_j b_j^x X_j^h$$

- a_i^z, b_j^z: classical longitudinal biases
- W_{ij}: inter-layer ZZ coupling (classical part)
- a_i^x, b_j^x: **transverse fields** (quantum)
- No intra-layer ZZ couplings $\Rightarrow$ conditional independence preserved



Classical limit

Set $a_i^x = b_j^x = 0$: recovers exact classical RBM.

Self-loops denote transverse fields a_i^x, b_j^x (quantum fluctuations within each unit).

RQBM Gradient and Contrastive Divergence

For the RQBM, the gradient of the log-likelihood w.r.t. the inter-layer coupling W_{ij} is:

$$\frac{\partial \ln p(\mathbf{v})}{\partial W_{ij}} = \beta \left[\langle Z_i^v Z_j^h \rangle_{\text{clamp}} - \langle Z_i^v Z_j^h \rangle_{\text{free}} \right]$$

For the transverse-field bias a_i^x:

$$\frac{\partial \ln p(\mathbf{v})}{\partial a_i^x} = \beta \left[\langle X_i^v \rangle_{\text{clamp}} - \langle X_i^v \rangle_{\text{free}} \right]$$

Key difference from classical: the quantum gradient involves $\langle X_i \rangle$ — an off-diagonal expectation value with no classical analogue.

Quantum CD algorithm

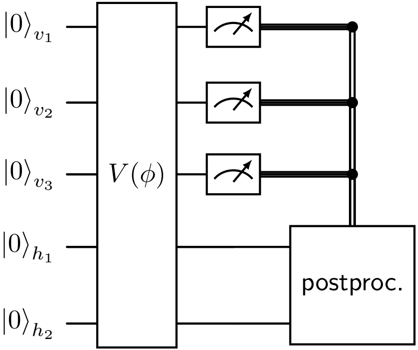
1. **Positive phase:** clamp v to data; sample h from $\rho_h^{(v)} = \mathrm{Tr}_v[\rho_{\text{clamp}}]$
2. **Negative phase:** run k steps of quantum Gibbs sampling (Metropolis or QITE) to approximate ρ_{free}
3. **Update:** $\Delta W_{ij} \propto \langle Z_i Z_j \rangle_+ - \langle Z_i Z_j \rangle_-$

Advantage over classical CD

Quantum sampling of $\langle X_i \rangle$ and $\langle Z_i Z_j \rangle$ is naturally unbiased on quantum hardware — no MCMC convergence issue.

RQBM Circuit: Sampling and Inference

Sampling from the RQBM: prepare the Gibbs state and measure the visible qubits.



Training (positive phase): clamp visible registers to data sample $\mathbf{v}$ and measure $\langle Z_i^v Z_j^h \rangle$, $\langle X_i^v \rangle$, $\langle X_j^h \rangle$.

Training (negative phase): run $V(\phi)$ freely; measure the same observables to estimate the model expectation values $\langle Z_i Z_j \rangle_{\text{free}}$.

Expressivity of QBMs vs. Classical BMs

What a classical RBM can represent

- Any distribution over $\{0,1\}^n$ with sufficiently many hidden units
- But: exponentially many hidden units needed to replicate some quantum distributions
- Limited to diagonal (classical) Hamiltonians

What a QBM adds

- Off-diagonal elements $\Rightarrow$ coherences
- Entangled marginals over visible qubits
- Quantum-correlated distributions not realisable by any classical latent variable model

Formal expressivity result: Any n-qubit ρ is the Gibbs state of some H:

$$\rho = \frac{e^{-\beta H}}{Z}, \quad H = -\beta^{-1}\ln(Z\rho)$$

$\Rightarrow$ QBM is a *universal* model for quantum states.

Sample complexity

Learning ρ to within ε requires:

$$N = O\left(\frac{n}{\varepsilon^2}\right) \text{ samples}$$

(quantum state tomography lower bound).

When Do QBMs Outperform Classical BMs?

Theoretical separation:

1. **Quantum data** (molecular wavefunctions, lattice gauge configs): QBM learns directly.
2. **Long-range entanglement:** exponentially many classical hidden units needed; $O(n)$ quantum hidden qubits suffice.
3. **Partition function estimation:** classical MC suffers sign problems for fermionic systems; QBM avoids this.

Practical regime (NISQ):

- Small n ($\lesssim 30$ qubits)
- Shallow thermal circuits
- Heuristic advantage — not yet provable

Caution: no-go results

For *classical* data distributions:

- Any QBM can be dequantised if quantum correlations of ρ are weak.
- Advantage requires genuinely quantum off-diagonal structure.

Barren plateaus in QBMs

Like QNNs, deep circuits suffer $\mathrm{Var}(\nabla_\theta \mathcal{L}) \sim 2^{-n}$.
Mitigation: shallow circuits and problem-inspired Hamiltonian structure.

Outlook: Open Problems and Future Directions

Near-term (NISQ) focus

- Small QBMs ($n \lesssim 30$) on current hardware
- Hybrid: quantum sampling + classical update
- Applications: quantum chemistry, finance, combinatorial optimisation
- Error mitigation for thermal state preparation

Key open questions

- Can QBMs learn classically hard-to-sample distributions efficiently?
- Can barren plateaus be avoided with problem-inspired structure?
- Quantum sample complexity advantage?

Long-term directions

- **Fault-tolerant QBMs:** exact Gibbs state via quantum signal processing
- **Quantum GANs:** replace discriminator or generator with a QBM
- **Physics:** learn Hamiltonians from data; phase diagrams; sign-problem-free MC
- **Tensor networks:** finite-T DMRG $\leftrightarrow$ QBM circuits

Quantum advantage landscape:

Classical BM $\subset$ QBM $\subset$ Full QC

Summary: QBM in Context

	Classical BM	QNN	QBM
State	$p(\mathbf{v})$	Pure $\|\psi(\theta)\rangle$	Mixed $\rho(\theta)$
Model	$e^{-\beta E}/Z$	$U(\theta)\|0\rangle$	$e^{-\beta H(\theta)}/Z$
Gradient	CD (biased)	Param-shift	Thermal avg.
Geometry	Euclidean	Fubini-Study QFI	Bures metric
Training	Gibbs sampling	TDVP (real time)	TDVP (imag. time)
Expressiv.	Classical dists.	Pure states	All quantum states
Advantage	Tractable	Discriminative	Generative

The central message

A QBM is a **thermal QNN**: it replaces the pure-state ansatz with a Gibbs state, enabling generative modelling of quantum distributions with a natural connection to statistical mechanics.

References

Quantum Boltzmann Machines

1. M. H. Amin et al., *Quantum Boltzmann machine*, Phys. Rev. X **8**, 021050 (2018).
2. G. Verdon et al., *A quantum algorithm to train neural networks using low-depth circuits*, arXiv:1712.05304 (2017).
3. J. Gao & L.-M. Duan, *Efficient representation of quantum many-body states with deep neural networks*, Nat. Commun. **8**, 662 (2017).
4. C. Zoufal et al., *Variational quantum Boltzmann machines*, Quantum Mach. Intell. **3**, 7 (2021).

Thermofield double and Gibbs state preparation

5. S. Chowdhury et al., *A variational quantum algorithm for preparing quantum Gibbs states*, arXiv:2002.00055 (2020).
6. D. Zhu et al., *Generation of thermofield double states and critical ground states with a quantum computer*, PNAS **117**, 25402 (2020).

QNN and QFI (see also week15)

7. J. Stokes et al., *Quantum natural gradient*, Quantum **4**, 269 (2020).
8. M. H. Amin, *Searching for quantum speedup in quasistatic quantum annealers*, Phys. Rev. A **92**, 052323 (2015).
9. R. Babbush et al., *Focus beyond quadratic speedups for error-corrected quantum advantage*, PRX Quantum **2**, 010103 (2021).

Course Philosophy and Structure

Two main parts

1. **Quantum Computing** (Jan–Mar)
   Many-body quantum systems, VQE, fault-tolerant algorithms
2. **Quantum Machine Learning** (Apr–May)
   NISQ-era variational methods, QNNs, QPINNs, Boltzmann machines

Central Algorithm

The **Variational Quantum Eigensolver (VQE)** links both halves of the course.

Key Themes

- Hilbert spaces & quantum states
- Entanglement & measurement
- Variational circuits (NISQ)
- Fault-tolerant algorithms
- Quantum-enhanced ML
- Solving ODEs/PDEs with quantum methods

Week 1 (Jan 19–23) Basic Notions of Quantum Mechanics

Mathematical Framework

- Linear algebra reminder
- Hilbert spaces and operators
- Bra-ket notation: $|\psi\rangle$, $\langle\phi|$
- Inner products & norms

Quantum States

- One-qubit systems
- Pure states vs. mixed states
- Composite systems
- Operator algebra on Hilbert spaces

Foundation

Establish the mathematical language used throughout the entire course.

Weeks 2–3 (Jan 26 – Feb 6) Composite Systems, Density Matrices & Measurements

Week 2: Tensor Products & Entanglement

- Spectral decomposition
- Measurement formalism
- Density matrices $\rho = \sum_i p_i |\psi_i\rangle \langle\psi_i|$
- Entanglement: pure & mixed states

Week 3: Density Matrices in Depth

- Reduced density matrices
- Partial traces
- Von Neumann entropy
- Entanglement measures

Key Concept: Entanglement

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big) \quad \Rightarrow \quad S(\rho_A) = \log 2$$

Week 4 (Feb 9–13) Entanglement, Entropies & Quantum Gates

Entanglement & Entropies

- Review of density matrices
- Entanglement entropies
- Separability criteria

Quantum Gates (Introduction)

- Single-qubit gates: X, Y, Z, H, S, T
- Two-qubit gates: CNOT, CZ, SWAP
- Gate universality
- Simple Hamiltonian systems

Bridge to VQE

Understanding gates and Hamiltonians prepares for the VQE algorithm.

Week 5 (Feb 16–20) Getting Started with the VQE Algorithm

Quantum Gates & Simple Algorithms

- Quantum circuit model
- Parameterised circuits (ansätze)
- Gate decompositions
- One- & two-qubit Hamiltonians

VQE Overview

$$E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\, H \,|\psi(\boldsymbol{\theta})\rangle \;\geq\; E_0$$

Minimise $E(\boldsymbol{\theta})$ over circuit parameters $\boldsymbol{\theta}$.

Project 1

Implement VQE for simple
one- and two-qubit Hamiltonians.

Key tasks:

- Design ansatz circuit
- Evaluate energy expectation
- Classical optimisation loop

Week 6 (Feb 23–27) Implementing VQE: Measurements & Gradients

Advanced VQE Topics

- **Parameter-shift rule** for gradients:

$$\partial_\theta E = \frac{E(\theta+\frac{\pi}{2})-E(\theta-\frac{\pi}{2})}{2}$$

- Adaptive VQE (ADAPT-VQE)
- Shot-noise and measurement strategies
- Simulation codes and debugging

Hamiltonians Explored

- One-qubit spin models
- Two-qubit interacting systems
- Mapping physics to qubits
- Introduction to the **Lipkin model**

Weeks 7–8 (Mar 2–13) VQE for the Lipkin Model & Jordan-Wigner

Week 7: Lipkin Model VQE

- Two-qubit Hamiltonian implementation
- Full VQE pipeline for the Lipkin collective model
- Benchmarking against exact diagonalisation
- Circuit optimisation strategies

Week 8: Fermionic Hamiltonians

- Lipkin model: final results
- **Jordan-Wigner transformation**:

$$a_j \;\rightarrow\; Z^{\otimes j} \otimes \sigma^-$$

- Mapping fermionic operators to qubits
- Beyond the Lipkin model

Milestone

Project 1 results in a working VQE solver.

Week 9 (Mar 16–20) Quantum Fourier Transforms

From NISQ to Fault-Tolerant Algorithms

- Discrete Fourier Transform review
- **Quantum Fourier Transform (QFT)**:

$$\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i jk/N}\,|k\rangle$$

- Circuit decomposition: $O(n^2)$ gates vs. $O(n\,2^n)$

QFT Circuit Structure

- Hadamard gates
- Controlled phase rotations R_k
- Bit-reversal permutation
- Exponential speedup over classical FFT

Weeks 10–11 (Mar 23 – Apr 10) QFT, Quantum Phase Estimation & HHL

Quantum Phase Estimation (QPE)

- QFT as subroutine
- Estimating eigenvalues of U:

$$U |u\rangle = e^{2\pi i\varphi} |u\rangle$$

- Circuit construction & precision
- Applications to chemistry & physics

HHL Algorithm (Linear Systems)

- Solving $A\boldsymbol{x} = \boldsymbol{b}$ on a quantum computer
- Uses QPE as subroutine
- Exponential speedup (sparse, well-conditioned A)
- Implementation and codes
- Limitations and caveats

Week 12 (Apr 13–17) HHL Algorithm: Codes & Implementation

Deep Dive into HHL

- Full circuit implementation
- Ancilla qubit rotation: eigenvalue inversion
- Amplitude encoding of $\boldsymbol{b}$
- Uncomputation and measurement
- Error analysis and success probability

HHL Complexity

$$\mathcal{O}\left(\log N \cdot \kappa^2 \cdot \frac{1}{\epsilon}\right)$$

vs. classical $\mathcal{O}(N^3)$

where $\kappa =$ condition number,
$\epsilon =$ precision,
$N =$ matrix dimension.

Transition to Quantum Machine Learning

With QFT, QPE and HHL covered, we now turn to the **NISQ-era QML** methods for the remainder of the course.

Week 13 (Apr 20–24) QAOA: Quantum Approximate Optimisation

QAOA Algorithm

- Combinatorial optimisation problems
- Alternating cost and mixer unitaries:

$$|\boldsymbol{\gamma}, \boldsymbol{\beta}\rangle = U_M(\beta_p)\, U_C(\gamma_p) \cdots U_M(\beta_1)\, U_C(\gamma_1)\, |+\rangle^{\otimes n}$$

- Classical outer-loop optimisation
- Depth-p approximation ratio

Implementation Focus

- Mapping problems to Ising Hamiltonians
- Circuit construction
- Gradient evaluation
- Benchmarking on toy instances

Connection

QAOA is the optimisation counterpart to VQE and serves as a gateway into Quantum Machine Learning methods.

Week 14 (Apr 27 – May 1) Quantum Machine Learning: Foundations

QML Basics

- What is QML? Taxonomy of approaches
- Data encoding strategies:
  amplitude, angle, IQP, etc.
- Parameterised quantum circuits as models
- Expressibility and entangling capacity
- Barren plateaus and trainability

Quantum Neural Networks

- QNN architecture overview
- Layered variational circuits
- Loss functions & gradients
- Links to classical deep learning
- Connections to QAOA

Weeks 15–16 (May 4–15) QPINNs: Solving Differential Equations

Physics-Informed QNNs

- Classical PINNs review
- **Quantum PINNs (QPINNs)**: replace neural network with a parameterised quantum circuit
- Encoding PDEs as loss functions
- Automatic differentiation on circuits

Applications & Implementation

- Solving ODEs and PDEs: Schrödinger, heat, Poisson
- Boundary condition encoding
- Gradient computation via parameter-shift rule
- Discussion of codes

Key Idea

QPINNs marry the physics-informed training paradigm with the expressive power of quantum circuits to solve differential equations on near-term hardware.

Week 17 (May 18–22) Quantum Boltzmann Machines

Quantum Boltzmann Machines (QBM)

- Classical RBM / BM recap
- Quantum generalisation:
  thermal states as model distributions
- Training via quantum log-likelihood gradients
- Transverse-field Hamiltonians as model parameters
- Restricted QBM (RQBM) architecture

Connections & Outlook

- Links to VQE and variational circuits
- Generative quantum models
- Applications in physics & optimisation
- Open research problems

Course Conclusion

QBMs illustrate how quantum statistical mechanics and machine learning merge

Course Road Map — Part 1: Quantum Computing

Weeks	Dates	Topics
1	Jan 19–23	Basic quantum mechanics, Hilbert spaces, qubits
2–3	Jan 26–Feb 6	Composite systems, density matrices, entanglement
4	Feb 9–13	Entanglement entropies, quantum gates
5–6	Feb 16–27	VQE: algorithm, measurements, gradients (Project 1)
7–8	Mar 2–13	VQE for Lipkin model, Jordan-Wigner transform
9	Mar 16–20	Quantum Fourier Transform
10–11	Mar 23–Apr 10	QPE and HHL algorithm
12	Apr 13–17	HHL codes and implementation

Course Road Map — Part 2: Quantum Machine Learning

Weeks	Dates	Topics
13	Apr 20–24	QAOA: quantum approximate optimisation
14	Apr 27–May 1	QML foundations, Quantum Neural Networks
15–16	May 4–15	QPINNs: solving differential equations
17	May 18–22	Quantum Boltzmann Machines and course summary

Overarching Goal

Equip participants with the theoretical foundations and practical implementation skills to design, analyse, and run quantum algorithms on near-term and fault-tolerant hardware.

Summary: What we have done

Part 1 – Quantum Computing

- Formalism of quantum mechanics
- Entanglement & density matrices
- Variational algorithms (VQE)
- Fault-tolerant algorithms: QFT, QPE, HHL
- Physical models: Lipkin, Pairing

Part 2 – Quantum ML

- QAOA for optimisation
- Quantum Neural Networks
- Physics-Informed QNNs (QPINNs)
- Quantum Boltzmann Machines
- Connections to classical ML

Overarching Goal

Equip course participants with the theoretical foundations and practical implementation skills to design, analyse, and run quantum algorithms on near-term and fault-tolerant hardware.

Perspectives

Future Directions in Quantum Computing

Algorithms, Hardware, and Scientific Applications

Perspectives — Algorithms: NISQ and Fault-Tolerant

Near-Term (NISQ) Algorithms

- **Improved VQE ansätze:** hardware-efficient, UCC, ADAPT-VQE, symmetry-preserving circuits
- **Quantum error mitigation:** zero-noise extrapolation, probabilistic error cancellation
- **Barren plateau remedies:** layerwise training, local cost functions, better initialisation
- **Advantage benchmarks:** tasks where NISQ devices outperform classical methods

Fault-Tolerant Algorithms

- **Quantum simulation:** Trotterisation, qubitisation, linear combination of unitaries (LCU)
- **Improved QPE:** randomised and Bayesian phase estimation with fewer qubits
- **Beyond HHL:** quantum linear algebra (QSVM, quantum PCA), dequantisation
- **Quantum walks & search:** Grover speedups, walk-based graph algorithms, Monte Carlo

Perspectives — Applications in the Physical Sciences

Quantum Chemistry and Nuclear Physics

- **Electronic structure:** VQE and QPE for ground and excited states beyond coupled-cluster reach
- **Nuclear structure:** ab initio calculations using Lipkin, pairing, and shell-model Hamiltonians
- **Quantum advantage:** fault-tolerant simulation of strongly correlated systems (e.g. FeMoco)
- **Imaginary-time evolution:** thermal states and finite-temperature properties on quantum hardware

Condensed Matter and High-Energy Physics

- **Lattice gauge theories:** quantum simulation of QED, QCD, and the Hubbard model
- **Topological phases:** preparation and detection of topological order and anyons
- **Quantum gravity:** holographic duality (AdS/CFT) encoded on quantum circuits
- **Quantum sensing:** Heisenberg-limited parameter estimation, magnetometry, gravimetry

Perspectives — Quantum ML, Hardware, and Open Challenges

Quantum Machine Learning Frontiers

- **Quantum generative models:** quantum GANs, quantum diffusion, and QBMs beyond the RQBM
- **Quantum reinforcement learning:** RL on quantum hardware for quantum control and agentic workflows
- **Classical–quantum hybrid:** tensor networks and classical circuit simulation as a guide
- **Certification:** classical shadow tomography and cross-entropy benchmarking

Hardware and Open Challenges

- **Qubit platforms:** superconducting, trapped-ion, photonic, neutral-atom arrays
- **Error correction:** surface and color codes; logical fault-tolerant qubits
- **Scalability:** how many logical qubits for quantum advantage on real-world problems?

Key Insight

Quantum advantage requires both better algorithms *and* better hardware in tandem.